

Possible but Not Easy

The gap between theoretical and practical manipulation in voting

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BACKGROUND

Democracy runs on the assumption that elections reflect the genuine will of the people — an assumption that depends entirely on manipulation being either too computationally complex or too informationally demanding to be realistic.

The academic literature on computational voting theory and the empirical record on polling accuracy, read together, reveal a narrower gap than either field alone would suggest — and a democratic habit of demanding more preference data may be narrowing it further.

THESIS

While election manipulation is computationally straightforward in the voting systems democracies actually use, it is held in check by the unreliability of voter preference data — yet the public's growing demand for that same polling data, despite its inaccuracy, may be closing the gap between theoretical and practical manipulability.

Manipulation is more accessible than assumed

NP-hardness applies only to Second-order Copeland — used in no real democracy. Every common system (Plurality, Borda, Maximin) falls within polynomial time, making optimal manipulation theoretically efficient.

Bartholdi et al. (1989)

Modern conditions make it even more feasible

Single-peaked preferences — the norm in polarized two-party systems — make manipulation algorithms more tractable. Social media provides the delivery mechanism to execute preference-shifting at scale.

Auletta et al. (2023) · Givi et al. (2024)

The real barrier is informational, not computational

Polls claim 95% confidence while achieving ~60% accuracy. Undecided voters — the most targetable group — are the least accurately modeled. Bias peaks when data is most available and least reliable.

Counts (2020) · Bon et al. (2019) · Panagopoulos & Farrer (2014) · Hillygus (2011)

Real elections have structural resistance — but it is narrow

Robustness scores show most outcomes require more preference swaps than a manipulator could engineer. Resilience is highest in less polarized, proportionally represented electorates — conditions that describe fewer elections than we would like.

Boehmer et al. (2022) · Givi et al. (2024)

Public demand for polling data is closing the gap

Each election cycle produces more publicly available preference data driven by insatiable media and voter demand. As voters broadcast preferences openly through polls and social media, the electorate becomes more legible and the informational barrier thins. Undecided voters remain the most sought-after and least understood group — the sharpest irony in the entire system.

Hillygus (2011) · Auletta et al. (2023) · Bon et al. (2019)

CONCLUSION

The barrier to manipulation is not computational but epistemic. Electoral reform focused on voting system design addresses the wrong layer of the problem. The vulnerability is not in how votes are counted — it is in how preferences are known. The democratic impulse toward transparency may be the very mechanism through which manipulation becomes increasingly feasible. Democracy runs on assumptions — and some of those assumptions are growing less secure.

SELECTED SOURCES

- Bartholdi, J. J., et al. *Social Choice and Welfare*, 1989.
- Auletta, V., et al. *arXiv*, Cornell University, 2023.
- Boehmer, N., et al. *arXiv*, 2022.
- Bon, J. J., et al. *Journal of the Royal Statistical Society*, 2019.
- Counts, L. *UC Berkeley Research*, 2020.
- Givi, G. M., et al. *Scientific Reports*, 2024.
- Hillygus, D. S. *Public Opinion Quarterly*, 2011.
- Panagopoulos, C. & Farrer, B. *Presidential Studies Quarterly*, 2014.

